FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO

# Feature Selection Optimization through Genetic Programming in Machine Learning Models

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# Chapter 1

# Introduction

## 1.1 Context

Datasets used to train machine learning (ML) models frequently contain dozens, hundreds, or even thousands of features, although only part of them actually carries information about the target variable. Some application areas take this to the extreme. In the quantitative structure–activity relationship (QSAR) benchmarks studied by Wang et al. [11], which relate chemical compounds to their activity on drug targets, each sample is described by 1024 variables, while fewer than 700 samples are available. With so many candidate inputs, a model can easily latch onto misleading patterns, training becomes more expensive, and explaining how the model reaches its predictions becomes harder [1][2].

Feature selection addresses this issue by discarding the inputs that do not help and keeping those that do. When it is performed well, predictors can become more accurate, cheaper to train and to run, and easier to understand [2]. The literature usually distinguishes three ways of performing it [2][12]. *Filters* rate variables through statistical criteria computed from the data alone, before any model is involved, whereas *wrappers* repeatedly train a chosen model on different candidate subsets and keep the one under which it performs best. *Embedded* methods, in turn, choose variables while the model itself is being trained. In many practical pipelines, the selection relies on per-feature importance scores, which are nowadays often obtained from explainability tools such as SHapley Additive exPlanations (SHAP) [3][4][10] and Local Interpretable Model-agnostic Explanations (LIME) [8].

Evolutionary computation (EC) approaches the selection problem as a search. In a Genetic Algorithm (GA), each candidate solution can be represented as a string of bits stating which variables are kept, which makes encoding a subset straightforward [12][14]. Genetic Programming (GP) goes one step further, since it evolves the predictive model itself in the form of an expression tree, and therefore only the variables that end up inside the evolved expressions are effectively used [12][14]. As a result, selection and model construction happen within a single process, and the resulting expressions can be read and inspected [6]. Building on these ideas, recent studies have paired a GA, responsible for choosing the subset, with GP, responsible for learning from

it [9], while others have used Shapley values to decide which variables a GP system should retain [11]. Within this context, the present dissertation, carried out with Altice Labs, investigates how such evolutionary strategies can deliver feature selection for ML models that is both more effective and less costly.

## 1.2 Motivation

Ranking variables one by one assumes that the usefulness of each variable can be judged in isolation, an assumption that often fails in practice. Guyon and Elisseeff, for example, describe cases in which a variable with no predictive power on its own becomes informative once paired with another, as well as cases in which strongly correlated variables still add information to each other [2]. Similarly, permutation importance, which perturbs a single variable at a time, is known to misjudge variables that are correlated with others [11]. Consequently, when such rankings drive the selection, variables that only matter jointly risk being discarded, and the final model may lose both accuracy and credibility.

GP is appealing in this setting precisely because the expressions it evolves can combine variables explicitly. However, its implicit selection loses effectiveness as the number of candidate variables grows [1][11], and repeated runs may settle on different subsets [7]. Shapley values offer a principled way of measuring the contribution of each variable, because they average its effect over the possible combinations of the remaining variables [3]. Indeed, using them to filter the inputs of GP improved its generalisation and reduced the number of selected variables by at least 27% relative to standard GP [11]. Nevertheless, the main obstacle remains the computational cost. Since Shapley values require evaluating each variable across many combinations of the others, the selection stage itself becomes expensive, and on one of the QSAR datasets mentioned above the Shapley-based pipeline took roughly five times longer than standard GP [11]. Nesting complete GP runs inside a GA is also demanding, given that every candidate subset triggers a full evolutionary search [9]. Reducing this cost without losing the ability to capture interacting variables would therefore benefit anyone who needs small, accurate, and understandable models, such as data scientists working with wide datasets in industry.

## 1.3 Problem

Feature selection currently involves a compromise between efficiency and the ability to account for interactions between variables. On the one hand, ranking-based methods are fast, yet they can discard variables whose contribution only becomes visible in combination with others [2][11]. On the other hand, the evolutionary alternatives that do account for such interactions are expensive, either because they repeat a full GP search for every candidate subset [9] or because they compute Shapley values in a dedicated and time-consuming phase before learning [11]. Moreover, work combining Shapley values with GP has concentrated on symbolic regression, a setting in which the evolved expression is itself the predictor [5][11], and the same holds for recent attempts to

feed importance information into the evolutionary operators [13]. It therefore remains unclear how well these ideas carry over to classification and whether the subsets they produce also benefit other kinds of ML models. In fact, Wang et al. themselves identify the integration of the Shapley computation into the evolution, the reduction of its cost, and its application to classification as directions for further work [11].

Resolving this compromise would be valuable in practice for teams that must reduce high-dimensional data under limited computing resources, since a selection procedure that keeps jointly relevant variables without an excessive runtime would lead to models that are smaller, more accurate, and easier to interpret.

## 1.4 Research Questions

Based on the gaps described above, this work is guided by the following research questions:

**RQ1** How can Shapley-value information be integrated into the GP evolutionary process, and how does this affect the computational cost of feature selection compared with two-stage and nested evolutionary approaches?

**RQ2** Does this integration preserve the quality of the selection, in terms of predictive performance, subset size, stability across runs, and the recovery of features that are only informative in combination, when compared with those approaches and with rankings based on individual feature importance?

**RQ3** To what extent are the selected subsets useful beyond GP itself, that is, when used to train other families of ML models, such as linear models, tree ensembles, and neural networks?

## 1.5 Hypothesis

Integrating Shapley-based importance into the GP evolutionary process reduces the computational cost of feature selection relative to two-stage and nested evolutionary approaches, while producing feature subsets that are at least as compact and accurate, and that remain useful when other families of ML models are trained on them.

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